

EEE 402 – Artificial Intelligence and Machine Learning Laboratory
2026 · Lab Group G2 · Group 13
Final Project Demonstration



TELLING LUNG ADENOCARCINOMA FROM SQUAMOUS CELL CARCINOMA ON CT

A radiomics model that still works on hospitals it has never seen

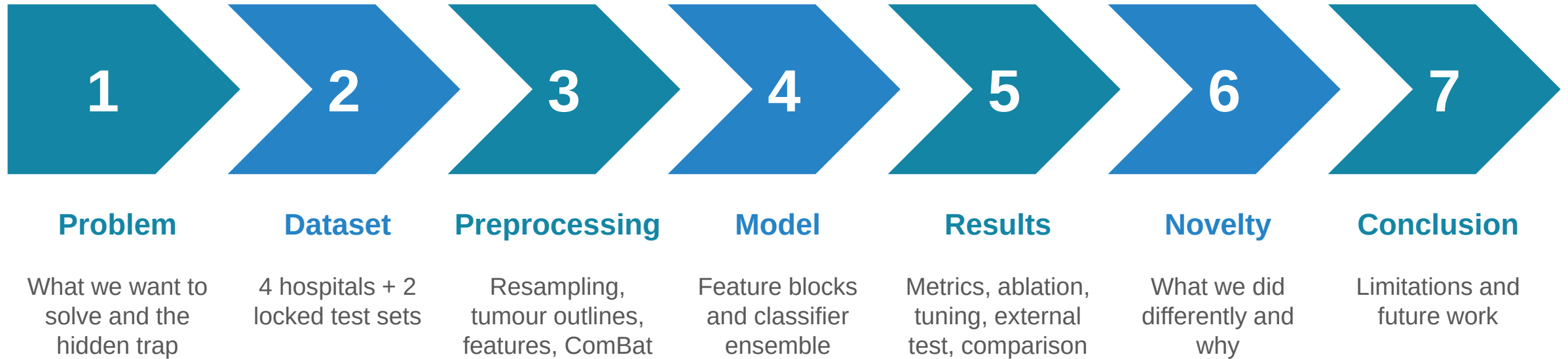
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Submitted by – Group 13, Lab Group G2



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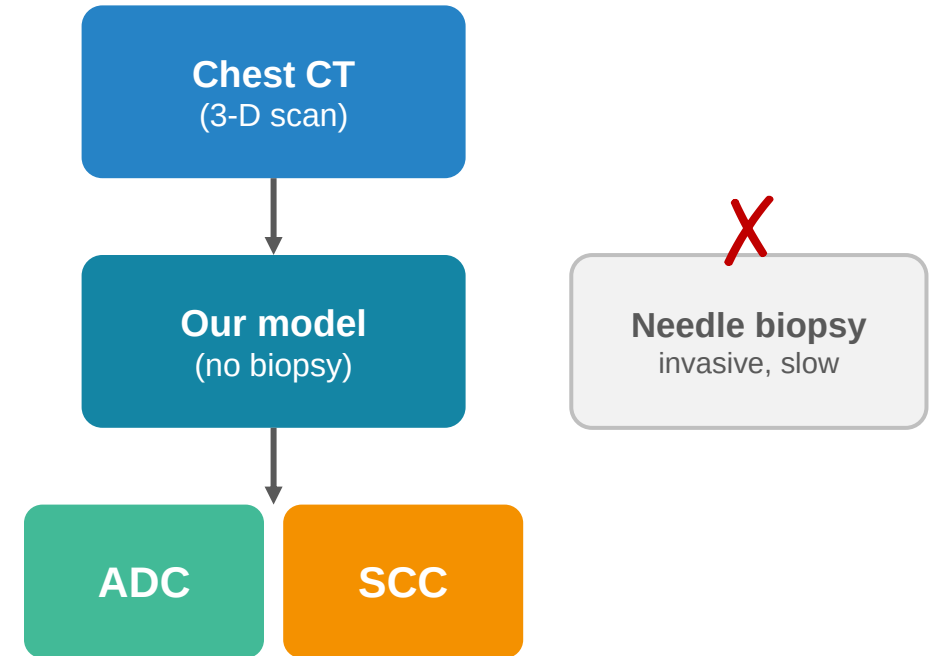
Outline



Aim: CT-based classification of ADC vs SCC, evaluated on hospitals not used for training and on two locked external test sets.

1. Problem Statement

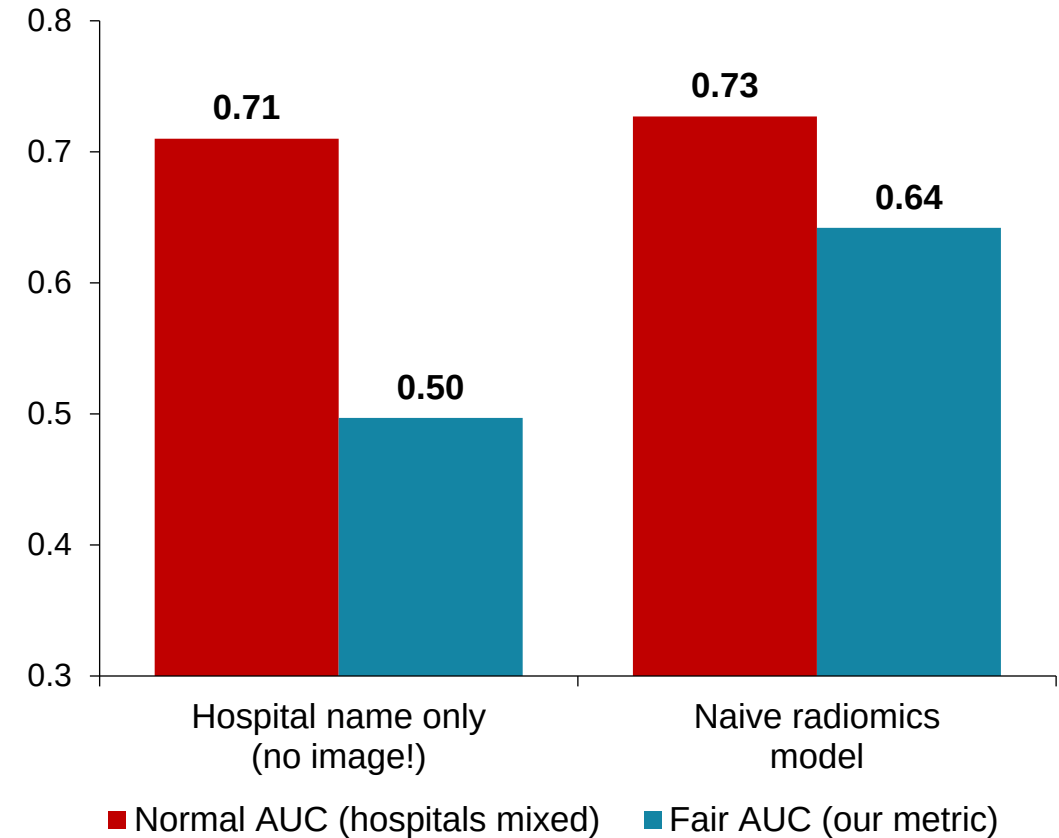
- Non-small cell lung cancer has two main types: **adenocarcinoma (ADC)** and **squamous cell carcinoma (SCC)**. Together they make up most lung cancer cases.
- They are treated differently – some chemotherapy drugs and targeted therapies are suitable for only one of the two types, so doctors must know the type before treatment starts.
- Today the type is confirmed by a **biopsy** (needle or surgery). It is invasive, has risks, takes time, and sometimes the sample is too small or cannot be taken at all.
- But almost every lung cancer patient already has a **CT scan**. So our question is: can a computer tell ADC from SCC just by looking at the CT?



Goal: predict ADC or SCC from a chest CT scan and the tumour location, with performance that holds on CT from new hospitals.

1.1 The Hidden Trap: the “Hospital Shortcut”

- Every hospital uses different scanners and settings, so its CT images carry a hidden “fingerprint” that a model can recognise very easily.
- Every hospital also has a different patient mix: **LUNG1 is 75% SCC**, while the other three hospitals are about **75% ADC**.
- So a model can learn “which hospital is this?” instead of “which cancer is this?” – and still get a high score when all hospitals are mixed together.
- We tested it: a model that only knows the **hospital name** and never looks at the image gets **AUC 0.71** on the normal metric!



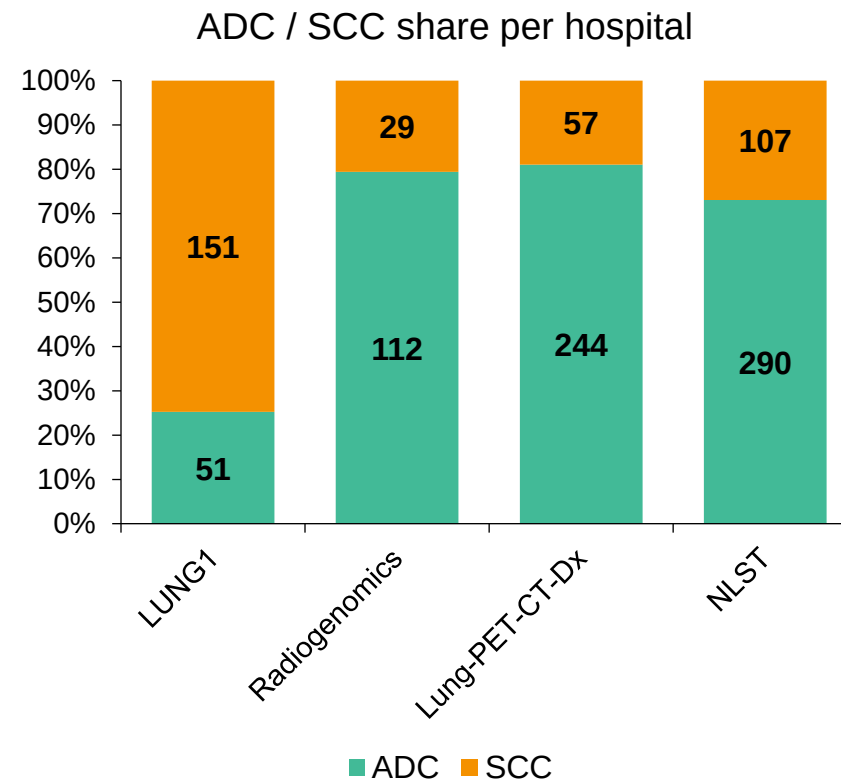
A high AUC on pooled multi-hospital data can reflect the hospital rather than the tumour. The 0.675 shown at the progress presentation was partly affected by this, so a fair metric is used from Section 5 onwards.

2. Dataset Description

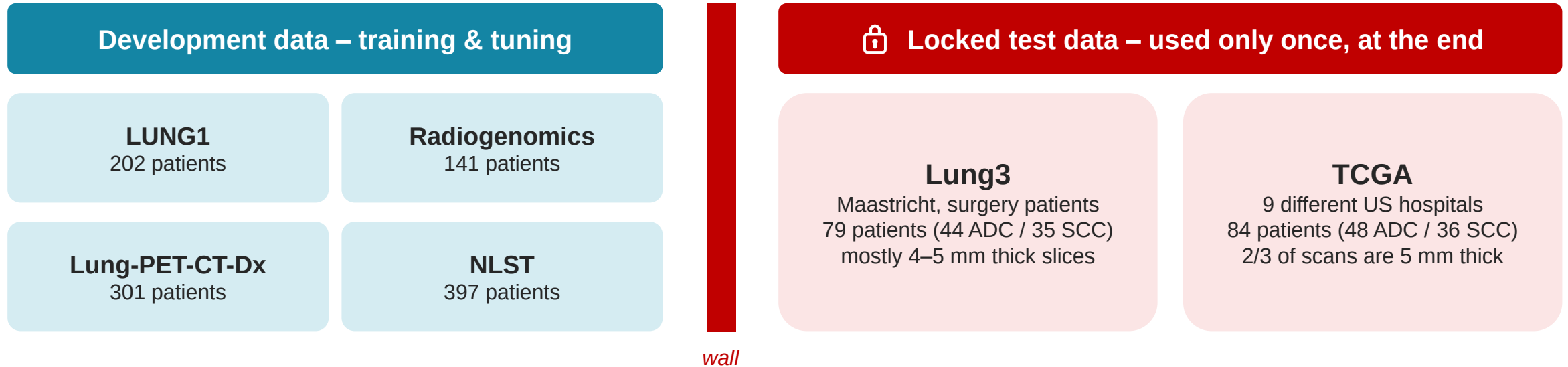
Dataset (hospital)	Country	Patients	ADC	SCC
LUNG1 (Maastro)	Netherlands	202	51	151
Radiogenomics (Stanford)	USA	141	112	29
Lung-PET-CT-Dx	China	301	244	57
NLST (screening trial)	USA	397	290	107
Total	3 countries	1041	697	344

- **Data type:** 3-D chest CT scans (DICOM, about 100–600 slices per scan), all public (TCIA and NCI IDC).
- **Classes:** 2 – ADC = 1 (positive class), SCC = 0. The label comes from the pathology report, one label per patient.
- **Tumour location:** expert outlines (LUNG1, Radiogenomics) or expert boxes (Lung-PET-CT-Dx, NLST).

LUNG1 is mostly SCC, while the other three hospitals are mostly ADC. This imbalance is what makes the hospital shortcut possible.



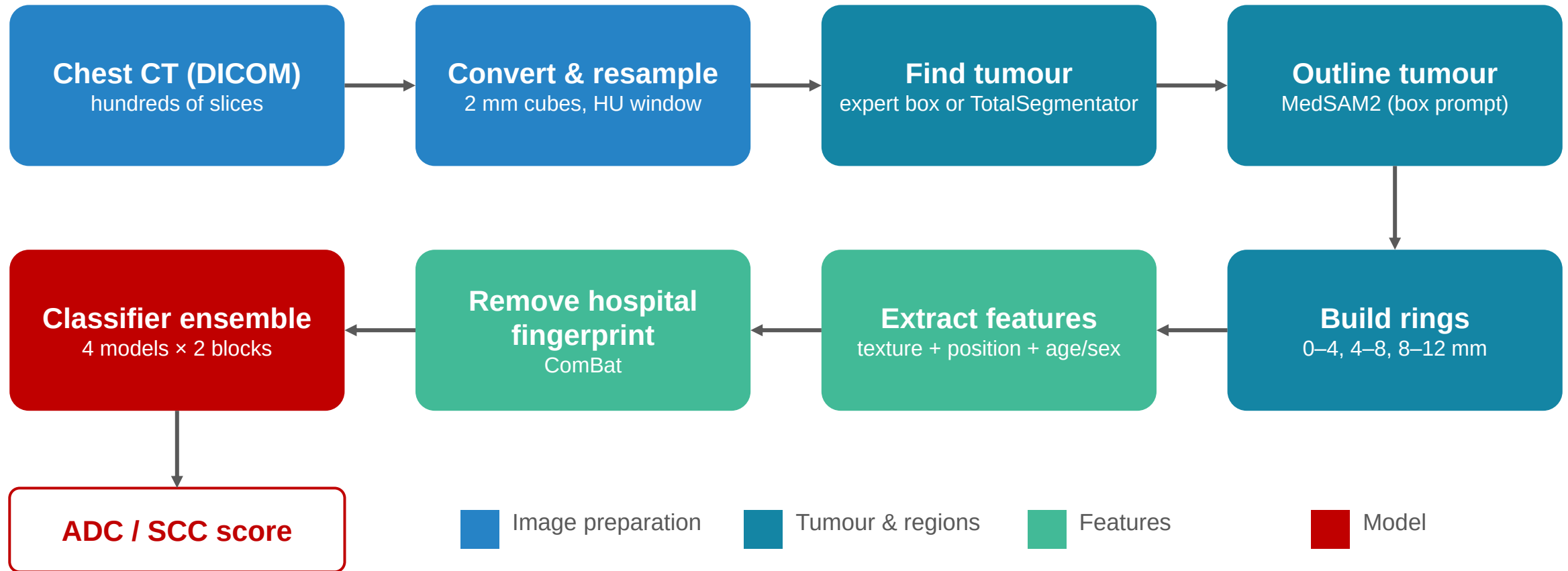
2.1 External Test Sets (Kept Locked)



- These two sets were kept completely separate: we never trained, tuned or even looked at their labels until the very end.
- The tumour was found **automatically** (TotalSegmentator + MedSAM2) – no human drew anything, exactly like a new hospital would use the model.
- Before opening the answers we wrote the test plan down and saved a fingerprint (**SHA-256 hash**) of all predictions, so nothing could be changed afterwards. Each test was scored **only once**.

The external tests follow the way the model would be used at a new hospital: no retraining, no tuning and no manual input.

3. Full Pipeline at a Glance

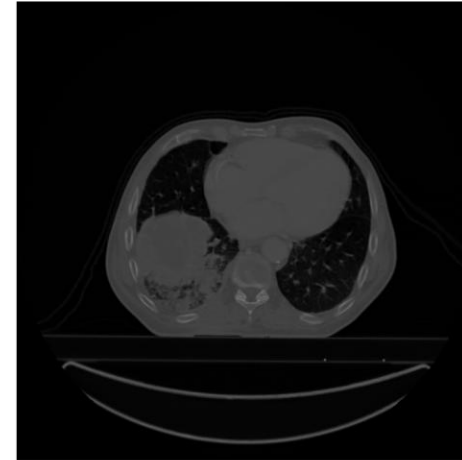


All steps are automatic. The same frozen pipeline was applied unchanged to both external test sets.

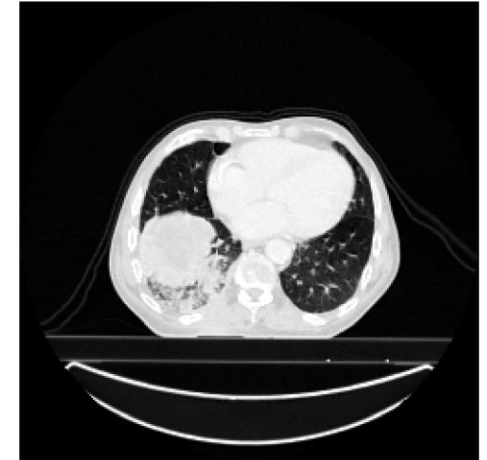
3.1 Preprocessing: Making Every Scan Look the Same

- Raw CT comes as hundreds of separate DICOM slice files. We convert each scan into **one 3-D volume** (NIfTI format).
- Hospitals use different pixel sizes (0.6–1 mm) and slice gaps (1–5 mm). We **resample everything to $2 \times 2 \times 2$ mm cubes** (linear interpolation for CT, nearest-neighbour for masks), so a 1 cm tumour means the same thing everywhere.
- CT values are in Hounsfield Units (HU). We keep **−1024 to 200 HU** (air to soft tissue), which removes bone and metal that could confuse the texture features.
- Grey levels are grouped into **20 HU bins** before texture is calculated, so small noise has less effect.

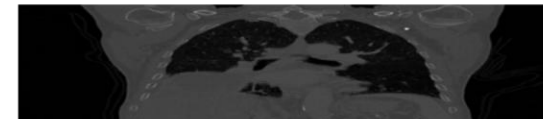
BEFORE – axial
0.98×0.98 mm pixels, full HU range



AFTER – axial
2×2 mm pixels, HU clipped to [−1024, 200]



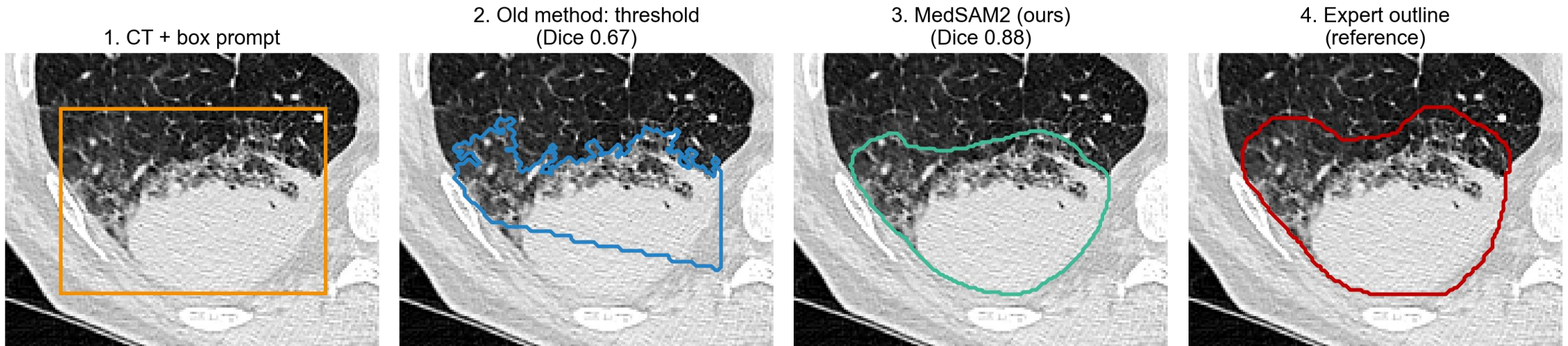
BEFORE – front view
3.0 mm between slices (squashed)



AFTER – front view
2 mm cubes in every direction

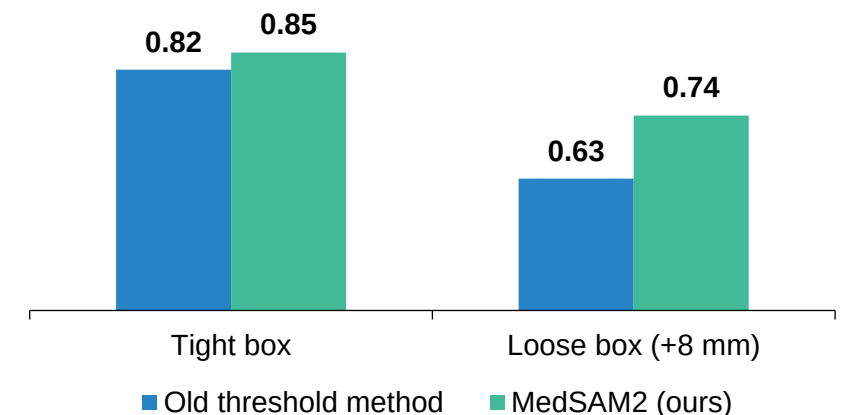


3.2 Preprocessing: Finding and Outlining the Tumour

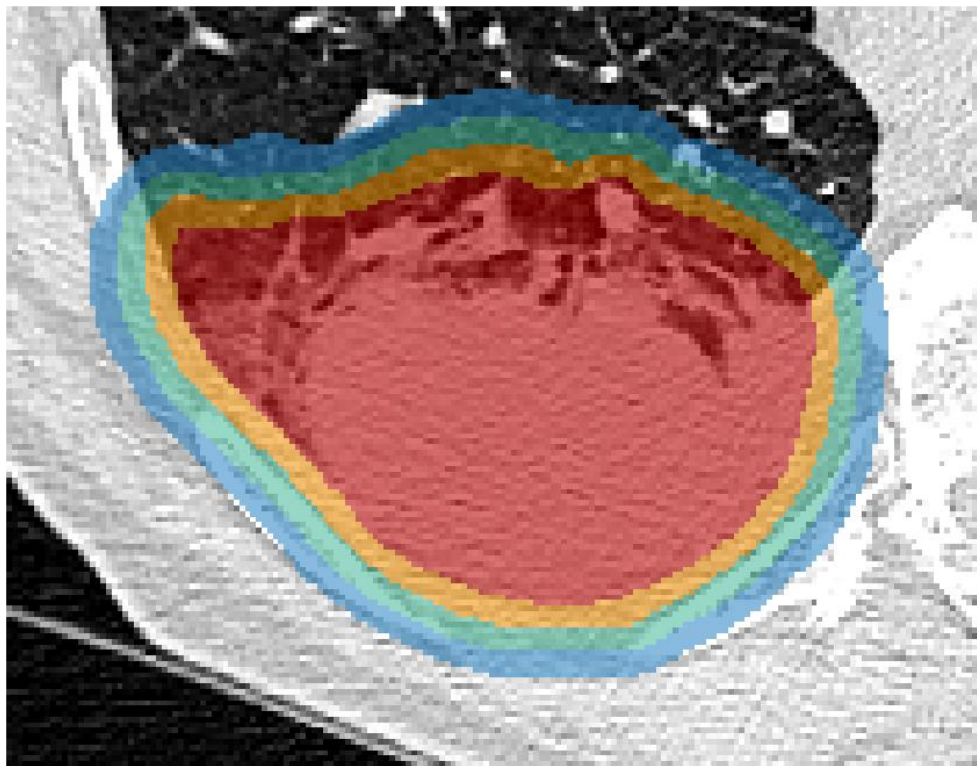


- Texture features need an exact tumour outline, but drawing outlines by hand for 1000+ patients is not possible for us.
- We give **MedSAM2** (a medical version of Meta's “Segment Anything” model) one box around the tumour. It outlines the tumour and follows it slice by slice through the 3-D scan.
- For new hospitals where no box exists, **TotalSegmentator** finds the lung nodule first – so the whole process is automatic.

Overlap with expert outline (Dice, 1 = perfect)



3.3 Features: Looking Inside and Around the Tumour



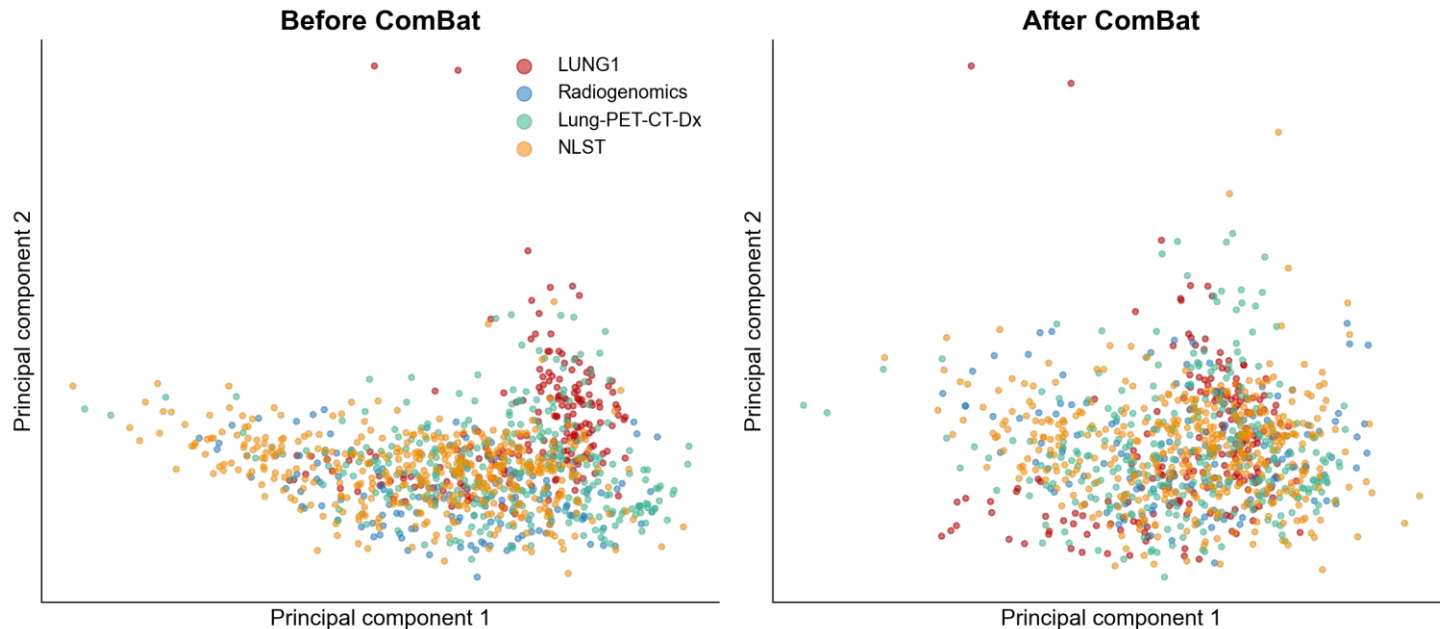
- Tumour (GTV)
-
- Ring 0–4 mm
-
- Ring 4–8 mm
-
- Ring 8–12 mm

- The tumour border carries information: ADC often grows along the air spaces with a fuzzy edge, while SCC is often more solid and central. So we look **inside and around** the tumour.
- We build **3 rings (shells)** outside the tumour – 0–4, 4–8 and 8–12 mm – and compute texture features in each ring with PyRadiomics.
- Texture families: first-order (brightness histogram), GLCM, GLRLM, GLSZM and NGTDM – they describe how grey levels repeat, form runs and form zones.
- **Position + clinical:** left/right lung, how central the tumour is, tumour volume, age and sex.

Feature block	What it describes	# features
Tumour texture (GTV)	inside the tumour	107
Rings (3 shells)	tumour border & surroundings	279
Position + clinical	where it is, size, age, sex	13



3.4 Removing the Hospital Fingerprint (ComBat)

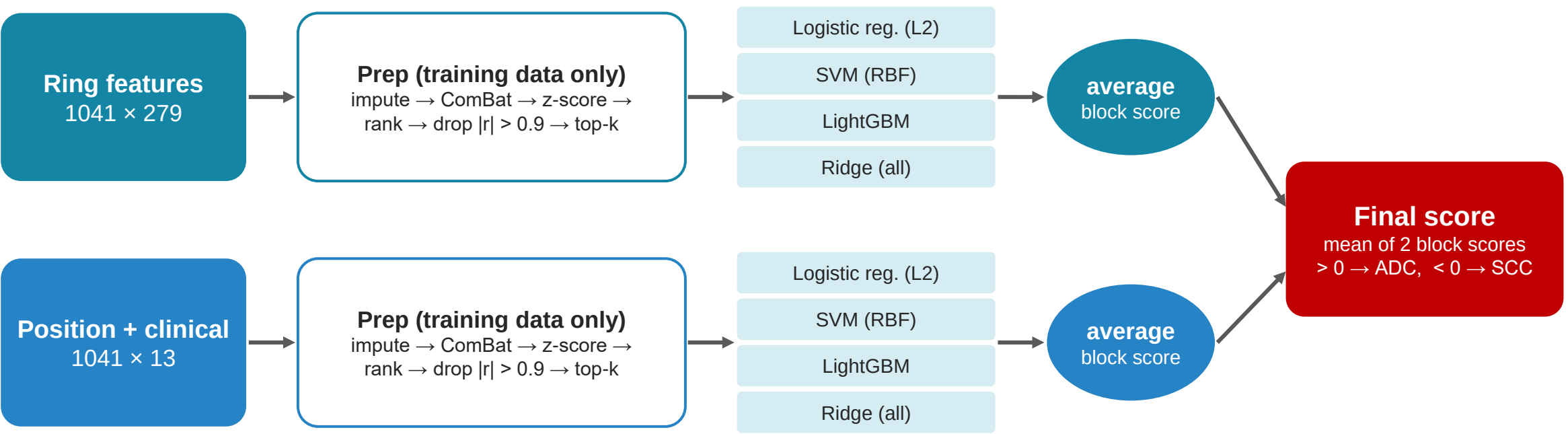


Each dot = one patient (2-D PCA of the ring features), coloured by hospital.

- Even after resampling, every hospital's features are shifted (different scanners and reconstruction). In the left plot, LUNG1 (red) forms its own cloud.
- **ComBat** (first made for gene-data batch effects) removes each hospital's own average shift and spread, feature by feature.
- It is fitted **only on training patients** inside each fold. A new hospital is corrected using its images only – no labels needed.
- Check: a “hospital detector” found the hospital with AUC ≈ 0.9 **before** ComBat and only $\approx 0.55\text{--}0.6$ **after**.

After ComBat the hospitals overlap, so the features reflect the tumour more than the scanner.

4. Model Architecture



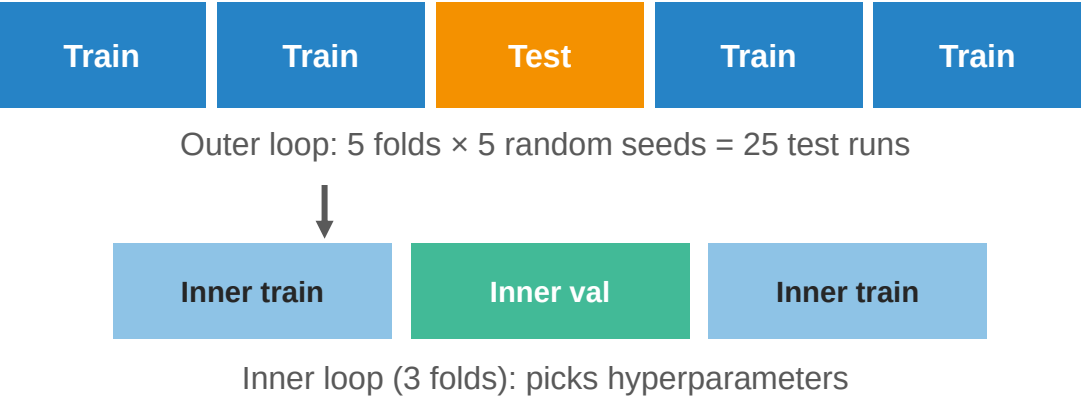
Item	Size / value
Input per patient	1 vector: 279 ring + 13 position numbers
Features kept per model (top-k)	10, 25 or 50 (tuned)
Output	1 number per patient (ADC score)

Why not a large CNN? With about 1000 patients, the deep models we tested (CNN and foundation models) scored lower and retained more hospital information (Section 5.4).



4.1 Training Setup and Hyperparameter Tuning

Nested cross-validation (patient level)



- Splits keep every hospital × class group in every fold.
- The test fold is **never** used for tuning or for ComBat.
- Sample weights make every hospital count equally and each 50/50 ADC/SCC (fixes class imbalance too).

Model	Hyperparameters tried	Most often chosen
Logistic reg. (L2)	C = 0.01 / 0.1 / 1; k = 10 / 25 / 50	C = 0.01
SVM (RBF)	C = 0.1 / 1 / 10; k	C = 0.1
LightGBM	leaves = 4 / 8; k (200 trees, lr 0.03)	4 leaves
Ridge (all features)	C = 0.0001 ... 0.01	C = 0.003–0.01
Selection metric	fair (site-balanced) AUC	–

Tuning result: the strongest regularisation (smallest C, fewest leaves) was selected in most folds, which indicates that simpler models generalise better on this data.

Test patients are never used for tuning, feature selection or harmonisation.



5. Results: How We Measure Fairly

① Site-balanced (fair) AUC



- Normal AUC mixes all patients; when hospitals have different ADC/SCC mixes, just knowing the hospital already “helps”.
- We re-weight patients (weight = $1 / \text{number in that hospital-and-class group}$), so **every hospital counts equally** and inside each hospital ADC and SCC count equally.
- A model that only guesses the hospital now scores **exactly 0.50**.

② Leave-one-hospital-out



- Train on 3 hospitals, test on the 4th, which the model has **never seen**. Repeat 4 times.
- This is the closest thing to “install the model in a new hospital”.
- Coloured = training hospitals, grey = unseen test hospital.

All results in this section are reported with these two methods, not with the standard pooled AUC.



5.1 Main Results

0.722

fair AUC on 4 hospitals
95% CI 0.69 – 0.76

0.714

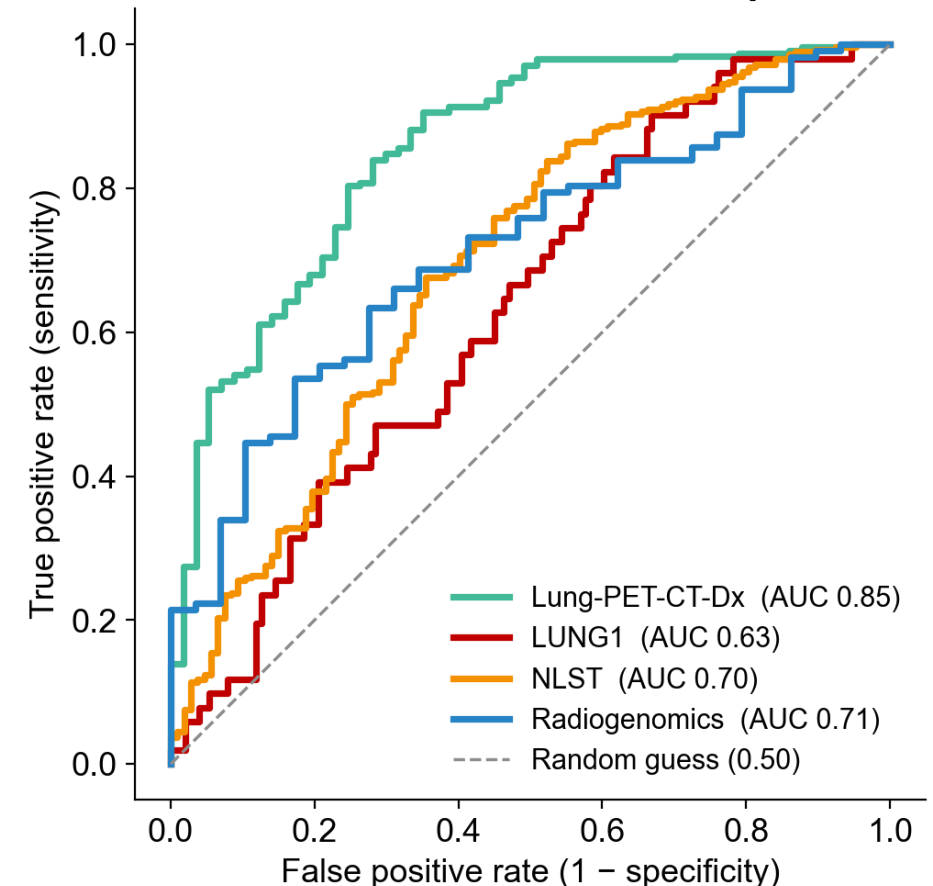
average AUC on an
unseen hospital

1041

patients from 4 hospitals
in 3 countries

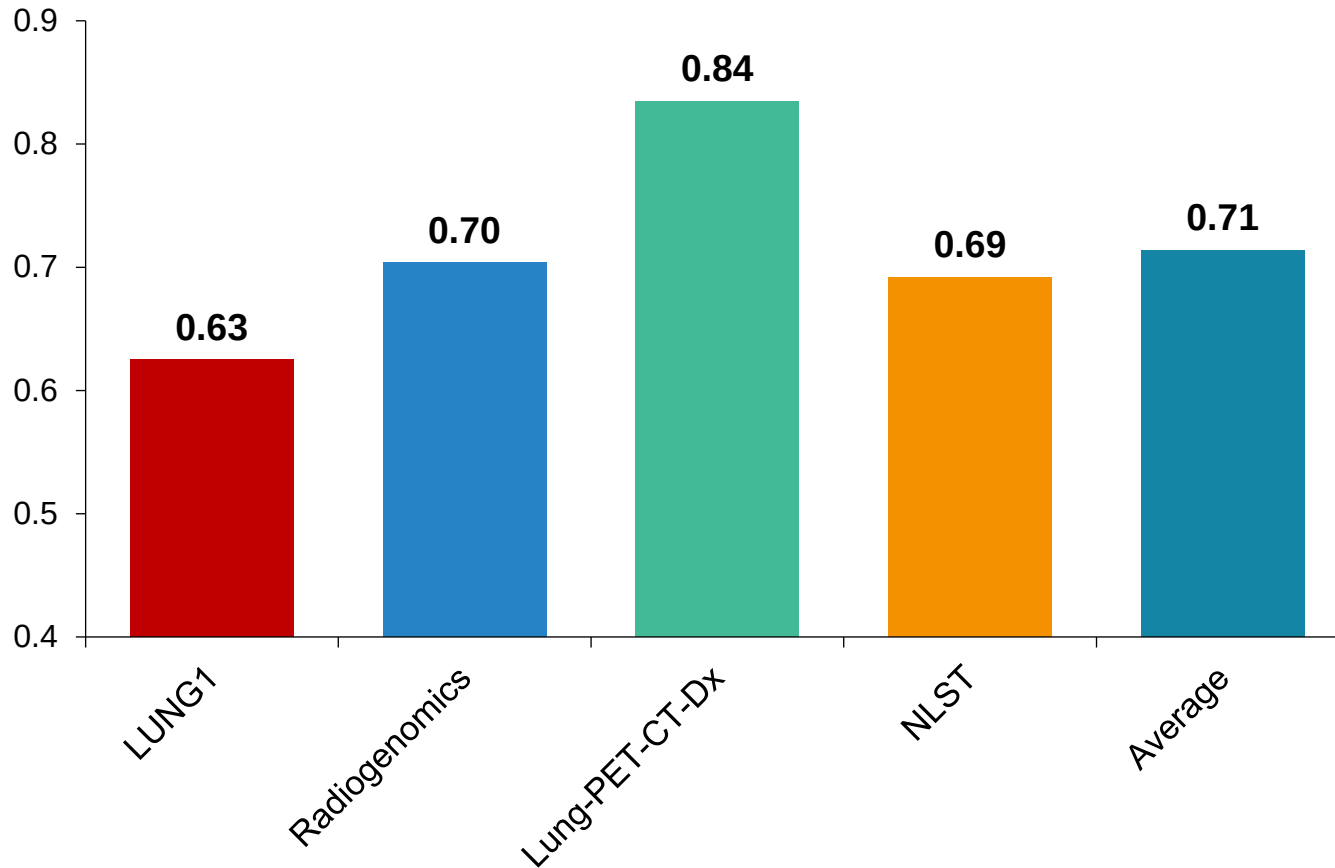
- All four hospitals are clearly above random (0.5). The best is **Lung-PET-CT-Dx (0.85)**, the hardest is **LUNG1 (0.63)**.
- LUNG1 is hard because 78% of its patients are late-stage (stage III), where ADC and SCC tumours look alike (similar size: 44 vs 48 ml).
- Every number is averaged over 5 seeds $\times$ 5 folds, so it is stable – not one lucky split.

ROC curve inside each hospital



The model performs above chance in every hospital under the site-balanced metric.

5.2 Result on Each Unseen Hospital

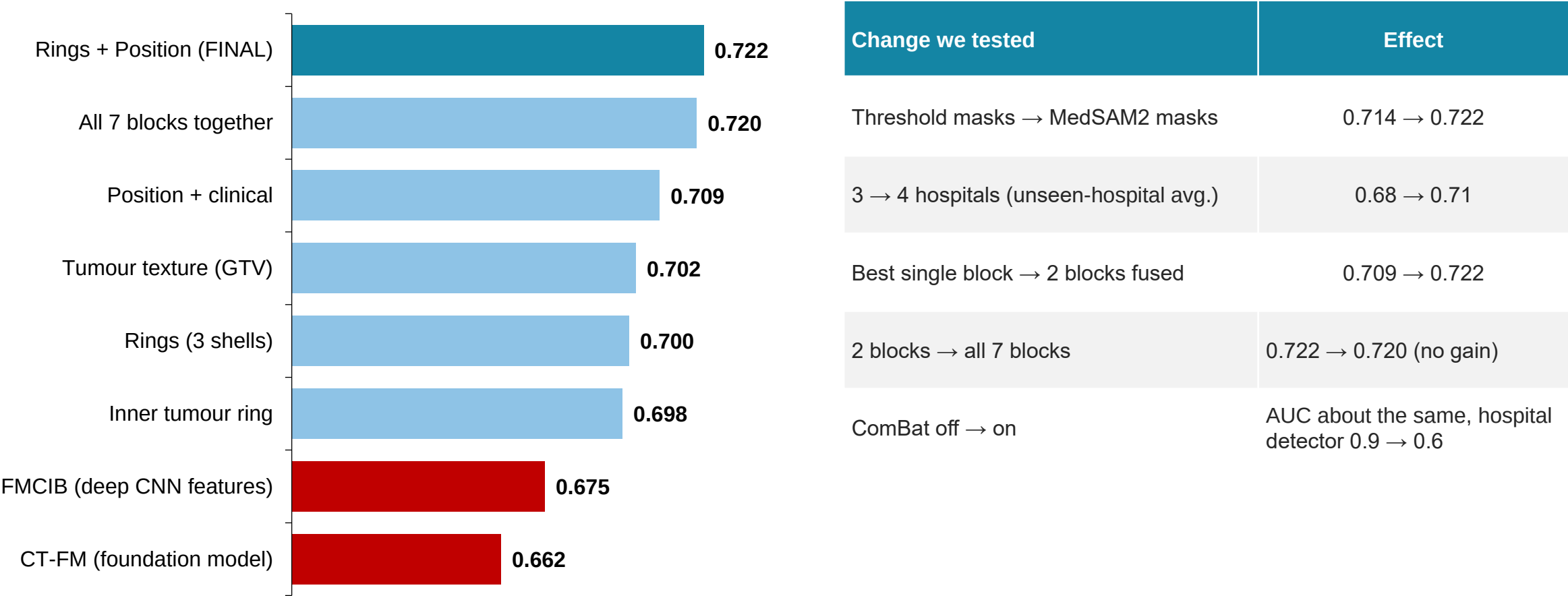


- Each bar: the model was trained on the **other 3 hospitals only**, then tested on this one.
- Unseen-hospital scores (0.63–0.84) are close to the pooled score (0.72), so the model is **not just memorising hospitals**.
- More hospitals helped: the unseen-hospital average went from **0.68 with 3 hospitals to 0.71 with 4**.
- Random guessing would be 0.50 for every bar.

Pooled fair AUC **0.722** vs unseen-hospital average
0.714
Difference of only 0.008

Leave-one-hospital-out gives the closest estimate of performance at a new hospital.

5.3 Ablation Study: What Matterred?



Accurate tumour masks and handcrafted ring and position features contributed more than deep features (red bars).

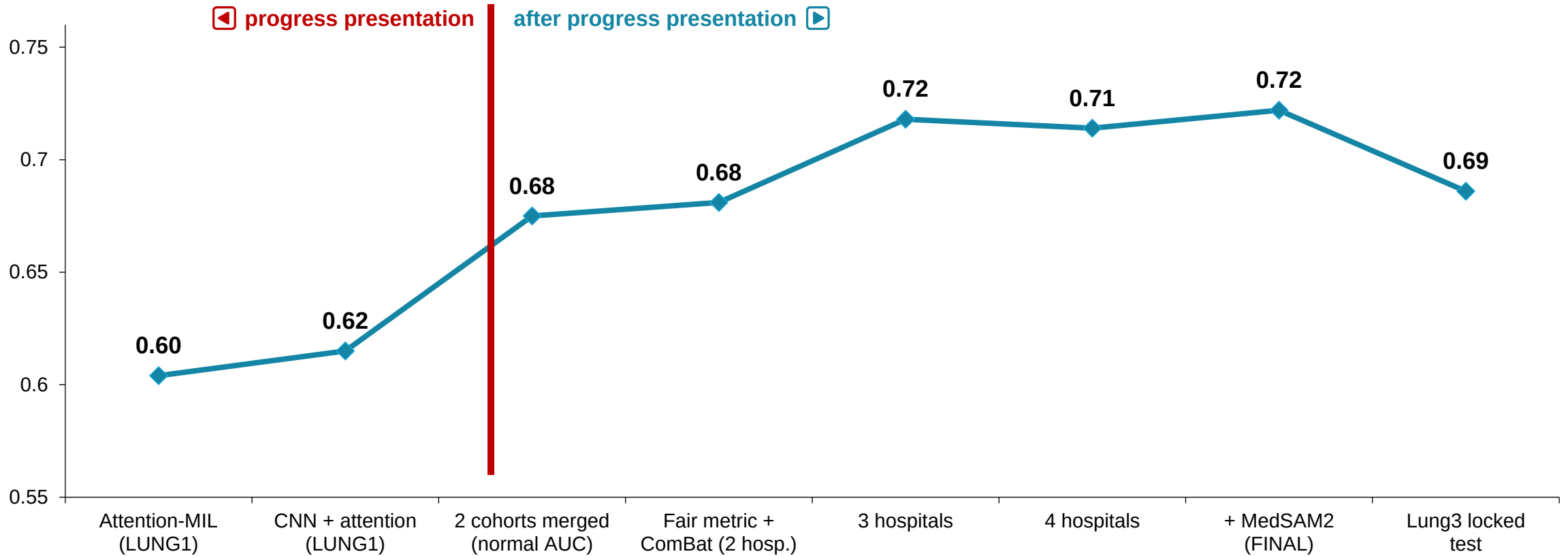


5.4 What Did NOT Work (and Why That Is Useful)

Idea we tried	Result (fair AUC)	What we learned
X Attention-MIL on small local patches (progress presentation)	≈ 0.60 (LUNG1)	Attention did not beat simple mean pooling; too little data
X Frozen deep CNN features (FMCIB)	0.675	Good, but weaker than handcrafted ring features
X CT foundation model (CT-FM)	0.662	Kept remembering the hospital, even after ComBat (detector 0.77)
X Fine-tuned CNN + “hospital-confusing” adversarial loss	0.606	Overfitted; learned the hospital even more (detector 0.82)
X Fusing all 7 feature blocks	0.720	No gain over the simple 2-block model

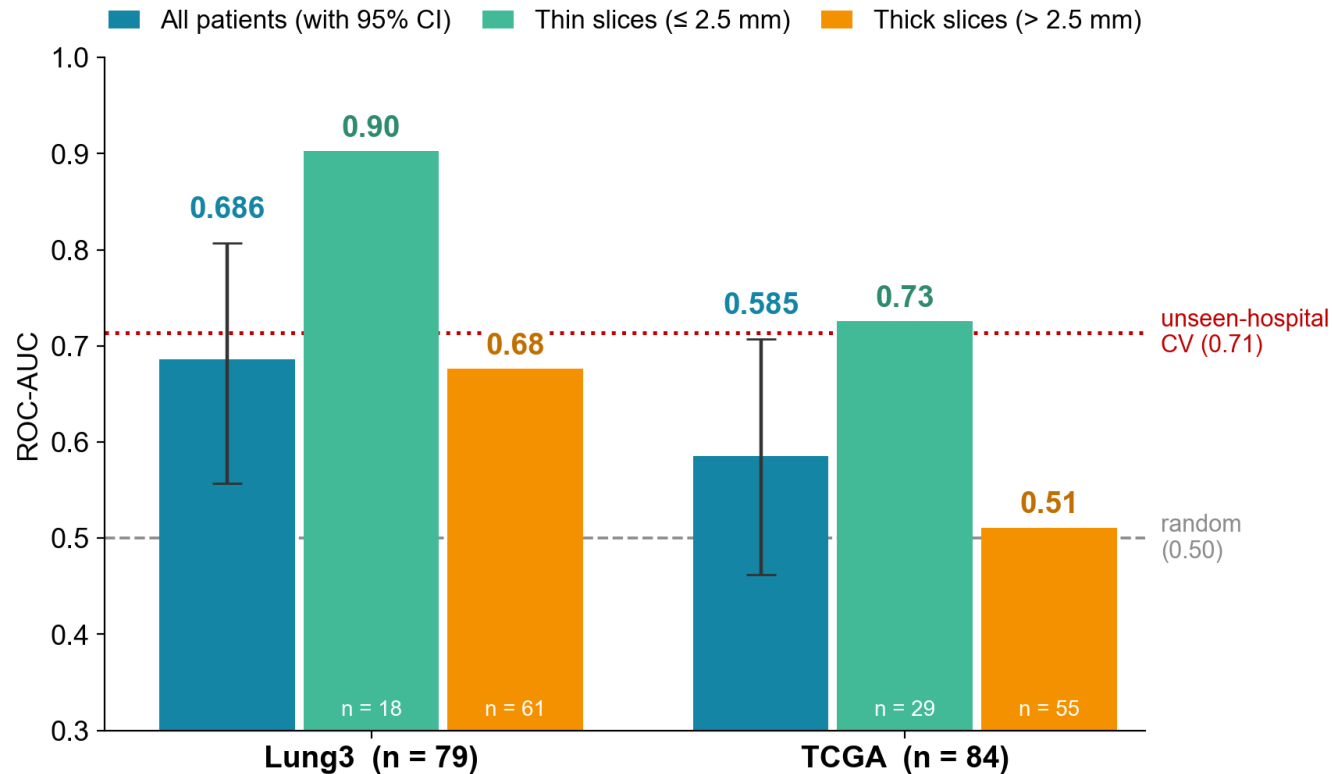
The larger deep models tended to learn hospital-specific patterns rather than tumour type, which limited their performance on this data.

5.5 Project Journey: From Progress Presentation to Final



The largest improvements came from adding hospitals, using a fair metric and improving the tumour masks. The 0.675 result (left of the red line) was partly inflated by the hospital shortcut.

5.6 Locked External Test on New Hospitals



- **Lung3: AUC 0.686** (sensitivity 0.64, specificity 0.69) – very close to our unseen-hospital CV (0.71). The model transferred.
- **TCGA: AUC 0.585** – lower. In TCGA each of the 9 hospitals sent **only ADC or only SCC** patients, so the test itself is mixed up with hospital effects (the trap from slide 4).
- **Thin-slice CT (≤ 2.5 mm) worked best** in both sets (0.90 and 0.73). Thick 5 mm slices blur the texture.
- TCGA tumours were also smaller (16 vs 29 ml), so automatic detection was harder.

Test protocol: the test plan was written in advance, predictions were saved with a SHA-256 hash before the labels were opened, and each test set was scored once. Error bars show 95% bootstrap confidence intervals.

5.7 Comparison with Existing Works

Study	Data	How it was tested	AUC	Remarks
Zhu et al. 2018, Eur Radiol	129 pts, 1 hospital	Split inside the same hospital	0.91	Small, single hospital, no external test
Pasini et al. 2023, Diagnostics	466 pts incl. LUNG1 + Radiogenomics	Random 80/20 split	Acc 0.77 → 0.59	Accuracy fell after ComBat – same hospital effect we found
Yang et al. 2021, Front Oncol	645 pts, 3 centres	Train 1 centre → test others	0.54–0.64	0.78 only when centres were mixed randomly
Chaunzwa et al. 2021, Sci Rep	311 pts, 1 hospital (MGH)	External: Lung3 (49 pts)	0.71 → 0.60	Deep CNN; same Lung3 test set as ours
Chen et al. 2023, Radiol Med	402 pts, TCIA	Internal / external (78 pts)	0.84 → 0.73	Multi-task CNN; clear drop on external
Song et al. 2023, Med Phys	868 pts, 8 TCIA sets	Internal + TCGA (97) + Lung3 (71)	0.82 / 0.82 / 0.80	Best of 130 models reported on these test sets; expert-drawn masks; hospitals mixed randomly
Ours (this project)	1041 pts, 4 hospitals	Unseen hospital + 2 locked tests	0.72 / 0.71 / Lung3 0.69 / TCGA 0.59	Fair metric, fully automatic masks, pre-registered locked tests

Most studies with AUC above 0.8 use one hospital or random splits; on independent hospitals, reported AUCs are 0.54 to 0.73. On Lung3 our model scored 0.69, compared with 0.60 (Chaunzwa et al.) and 0.80 (Song et al., expert masks, no locked test).

6. Novelty: What We Did Differently

1. Fair metric + fingerprint removal

What: site-balanced AUC and ComBat inside every fold.

Why: mixed-hospital AUC can be fooled by the hospital shortcut (0.71 with no image!).

Result: honest score rose from 0.64 (naive) to 0.72.

2. Fully automatic tumour outlining

What: TotalSegmentator finds the tumour, MedSAM2 outlines it from a single box.

Why: hand-drawing is impossible at a new hospital.

Result: Dice 0.85 vs experts; masks helped AUC 0.714 → 0.722.

3. 4 hospitals, 1041 patients

What: the largest public mixed-hospital set we know of for this task (3 countries).

Why: a model trained at one hospital does not travel.

Result: 0.71 on hospitals it never saw (0.68 with 3).

4. Locked, pre-registered external tests

What: plan written first, predictions hashed, scored once.

Why: no chance to “tune on the test set”.

Result: honest external numbers: Lung3 0.69, TCGA 0.59.

Summary: we first made the evaluation reliable across hospitals, and then improved the model under that evaluation.



7. Conclusion, Limitations and Future Work

Conclusion: a fully automatic CT pipeline classifies ADC vs SCC with a site-balanced AUC of 0.72 on four hospitals, 0.71 on unseen hospitals and 0.69 on the locked Lung3 test set.

Limitations

- Thick 5 mm CT slices hurt a lot (TCGA thick slices: 0.51).
- LUNG1 (late-stage patients) is still hard at about 0.63.
- External test sets are small (about 80 patients each), so their confidence intervals are wide.
- About 1 in 5 automatic detections picks the wrong nodule.

Future work

- Train with simulated thick slices so the model does not depend on slice thickness.
- Add PET scans, which carry metabolic information that CT does not.
- Test on a bigger external set with both classes from every hospital.
- Let a radiologist quickly confirm the detected nodule before prediction.



8. References

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- Pasini G. et al., “Phenotyping the histopathological subtypes of NSCLC: how beneficial is radiomics?,” Diagnostics, 2023.
- Yang F. et al., “Machine learning for histologic subtype classification of NSCLC: a retrospective multicenter radiomics study,” Front Oncol, 2021.
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- Wasserthal J. et al., “TotalSegmentator: robust segmentation of 104 anatomical structures in CT,” Radiol AI, 2023.
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- van Griethuysen J. et al., “Computational radiomics system to decode the radiographic phenotype (PyRadiomics),” Cancer Res, 2017.
- Datasets: TCIA NSCLC-Radiomics (LUNG1), NSCLC-Radiogenomics, Lung-PET-CT-Dx, NSCLC-Radiomics-Genomics (Lung3), TCGA-LUAD/LUSC; NCI IDC NLST.



Thank You!

Questions?

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